

Viraj Thakkar

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RESEARCH INTERESTS

Data Infrastructure: Key-Value stores (RocksDB); Timeseries databases (InfluxDB), Storage for Quantum Computing

Smart Storage: AI/ML for storage system; Real-Time Storage Systems for high-velocity and high-volume data

EDUCATION

Ph.D. Computer Science

Arizona State University, Tempe, AZ

Aug 2023 - Ongoing

3.91 / 4.00

M.S. Computer Science

Arizona State University, Tempe, AZ

Jan 2022 – Aug 2023

3.87 / 4.00

B. Tech. Information Technology

K. J. Somaiya College of Engineering, Mumbai, India

July 2017 – Aug 2021

8.56 / 10.0

PROJECTS

LLM-Assisted Tuning for LSM-KVS | C++, RocksDB, Docker, Python

12/2023 - Ongoing

- Lead a team of undergraduate and graduate students to design and develop a framework for automated prompt generation and feedback loop for generating configuration files for.
- Designed and Implemented architecture for multiple consistency levels within same database instance at the same time.

Smarter Consistency in Distributed KV-Stores | C++, HDFS, RocksDB, Kubernetes, Docker, ML

02/2022 – 08/2023

- Developed a scalable, distributed and storage-disaggregated Key-Value database service using RocksDB, HDFS, Docker, and Kubernetes.
- Designed and Implemented architecture for multiple consistency levels within same database instance at the same time.
- Currently implementing an ML model to automate assignment of consistency level to KV-pairs.

Memories: Image Segmentation Library | CI/CD, Python, OpenCV, PIL

08/2021 – 03/2022

- Creator and maintainer for the [‘Memories’](#) python library for Image Segmentation and Metadata management
- Achieved over 14,000 downloads till date.

Project Chakravayuh | DevOps (Azure, Docker), scikit-learn, pandas, OpenCV, Docker

09/2018 – 10/2020

- Led and developed a team of six who developed a compartmentalized system for dark-web crawling with capabilities of text-analysis, media-analysis, form-filling, website monitoring.
- Developed media-analysis module that tags any visual media based on its content using metadata and computer vision.
- Designed and implemented an automated form-filling module that facilitates access of content hidden behind login/sign-up prompts without user-interaction.

EXPERIENCE

Arizona State University: Graduate Service Assistant

08/2022 – 05/2023

- Grader for CSE-330 and CSE-511 for Fall 2022 and Spring 2023 semesters under Prof. Zhichao Cao.
- Implemented and assisted in re-designing course assignments and projects for CSE-511

Cyber Peace Foundation, Mumbai, IN: Analyst (Intern)

03/2020 - 08/2020

- Developed automated crawling and investigating pipeline in 2 months for abusive content on various social media.
- Team Leader for creation of detection and reporting tools for fake/malicious COVID websites. The tool successfully analyzed over 10,000 websites, identifying, and reporting 503 websites as malicious.

KEM Teaching Hospital, Mumbai, IN: Developer (Intern)

05/2020 - 08/2020

- Implemented and deployed an interactive dashboard for KEM Hospital to monitor patients in the COVID-19 pandemic.
- Tool used for 7+ months in the pandemic, assisting in management and cleaning 5000+ patient records to create a centralized hospital records database.

Somaiya Vidyavihar, IN & CIFF (Children’s Investment Fund Foundation), UK: Junior Researcher

09/2019 - 03/2020

- Worked on how technology has brought about Online Child Sexual Abuse and Exploitation (OCSA & OCSE).
- Developed tool to automate search and filter instances of OCSE and OCSE in Legal Cases

PUBLICATIONS

[SIGMOD '24] Qiaolin Yu, Chang Guo, Jay Zhuang, **Viraj Thakkar**, Jianguo Wang, Zhichao Cao. "CaaS-LSM: Compaction-as-a-Service for LSM-based Key-Value Stores in Storage Disaggregated Infrastructure". *Proceedings of ACM Conference on Management of Data (SIGMOD)*, 2024, To Appear.

[Book: RPA '23] Ashwini Dalvi, **Viraj Thakkar**, Smit Moradiya, Aditya Vedpathak, Irfan Siddavatam, Faruk Kazi, S. G. Bhirud. "FLASH: Web-Form's Logical Analysis & Session Handling Automatic Form Classification and Filling on Surface and Dark Web". *Robotic Process Automation (2023)*: 61-100. [URL](#)

[Book: HRAI-HCI '22] Irfan Siddavatam, Ashwini Dalvi, Abhishek Patel, Aditya Panchal, Aditya S. Vedpathak, **Viraj Thakkar**. "ML-Enabled Informed Intervention for Crowdsourcing-Based Optimization of Medical Resources." *Handbook of Research on Applied Intelligence for Health and Clinical Informatics*, pp. 291-315. IGI Global, 2022. [URL](#)

[ICACIT '21] Ashwini Dalvi, Irfan Siddavatam, **Viraj Thakkar**, Aditya Vedpathak, Abhishek Patel. "Recognizing Child Unsafe Apps Through User Reviews on the Google Play Store." *Advanced Computing and Intelligent Technologies* (pp. 111-120). Springer, Singapore 2022. [URL](#)

[ICAST '21] Irfan Siddavatam, Ashwini Dalvi, **Viraj Thakkar**, Aditya Vedpathak, Smit Moradiya, Apoorva Jain. "Explainability using decision trees and Monte Carlo simulations." *Proceedings of the 4th International Conference on Advances in Science & Technology (ICAST2021)*. [URL](#)

[IBSSC '21] Ashwini Dalvi, Irfan Siddavatam, **Viraj Thakkar**, Apoorva Jain, Faruk Kazi, Sunil Bhirud. "Link Harvesting on the Dark Web." *IEEE Bombay Section Signature Conference (IBSSC)*, pp. 1-5. IEEE, 2021. [URL](#)

ACADEMIC POSTERS AND WORK-IN-PROGRESS

[FAST '24] Madhumitha Sukumar, Jiaxin Dai, Kaushiki Singh, Vikriti Lokegaonkar, **Viraj Thakkar**, Zhichao Cao. "LLM-assisted Automatic-Configuration and Tuning Framework for LSM-based Key-Value Stores." *22th USENIX Conference on File and Storage Technologies*, 2024.

TECHNICAL SKILLS

Languages: Python, C/C++, Bash (Shell), SQL, MATLAB, Front End Development (HTML, CSS, JS)

Tools & More: Git, Kubernetes, Docker, Django, Pandas, TensorFlow, AWS, Azure, GCP, IBM Quantum

Databases & OS: PostgreSQL, RocksDB, Neo4j, MySQL, SQLite, Linux/Unix

Soft Skills: Project Management, Leadership, Consistency, Communication Skills, Creative problem solving.

AWARDS AND ACHIEVEMENTS

FAST '24 Travel Grant 2024

Stack Overflow 2023

- **1000+** Reputation on SO with 50+ answers. **Top 15%** contributor in 2023.

Hackathon by SVB & ASU, Tempe, AZ 04/2022

- Developed full stack web application to connect Investors with Startups. Achieved second place in the event.

Internship and Hackathon Organizer, Mumbai, IN 08/2020 - 04/2021

- Organized and coordinated a college-wide internship activity alongside college faculties. Created and managed task assignments and updated for 100+ interns over 2 semesters.
- Worked alongside college's computer science society for organizing and managing College Hackathon.

Team KJSCE Robocon, Mumbai, IN 08/2018 - 08/2020

- Spearheaded a team of 40+ individuals and mentored members to take over the team in coming years.
- Achieved National Rank 8 and Winner of Best Solution Idea amongst 50+ teams in "DD-Robocon 2020".